Project SRS Report (**AI Powered Personal Trainer**)

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Problem understanding & Analysis	7					
Implementation	8					
Report writing	10					
					Total obtained mark	
Comments						

Semester: Fall 2026

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Course Code: SE-331

Course Name: Software Engineering Design Capstone Project
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Submission Date: 11-10-26

1. Introduction

1.1 Problem Statement

Most people follow generic, non-personalized workout and diet routines that do not account for their individual physical characteristics, fitness level, goals, or physical limitations. Hiring a personal trainer for continuous, individualized guidance and real-time exercise-form correction is costly and not accessible to everyone. As a result, users often perform exercises with incorrect form without realizing it, which can reduce effectiveness and increase injury risk, and their workout/diet plans are rarely adapted based on their actual progress.

AI is needed here because generating a truly individualized plan requires analyzing multiple interacting variables per user (goals, body metrics, experience, equipment, limitations) rather than applying one fixed template, and because detecting exercise form from an image requires Computer Vision/Pose Estimation rather than manual rule entry.

1.2 Purpose

The primary objectives of this software are to:

- Collect individual user fitness and health-related data through a secure web platform.
- Use AI-assisted analysis to generate personalized workout and nutrition plans.
- Use a Computer Vision / Pose Estimation model (MediaPipe) combined with rule-based biomechanical analysis to evaluate exercise form from an uploaded photo (and, where feasible, video/live camera).
- Track user workout, nutrition, and progress history over time.
- Adapt future workout and nutrition recommendations based on the user's recorded progress.

1.3 Scope

What the system will do:

- User registration, login, and fitness-profile management.
- BMI, calorie, and protein calculation from user-provided data.
- Personalized, rule-based workout and nutrition plan generation.
- Photo-based exercise form detection using MediaPipe Pose and joint-angle rule analysis (core AI feature).
- Attempted video-based and live-camera-based form detection, where technically feasible.
- Workout, nutrition, and progress tracking with historical comparison.
- Adaptive recommendation updates based on recorded progress.

What the system (including its AI) will not do:

- Diagnose medical conditions or prescribe medication.
- Replace a doctor, physiotherapist, or certified personal trainer.

- Guarantee 100% accurate exercise-form detection under all camera angles and lighting conditions.
- Guarantee real-time live-camera accuracy under all environmental conditions.

2. Design and Implementation Constraints

2.1 Programming Language

Frontend: React.js with Vite (JavaScript/HTML/CSS) — chosen for fast, component-based UI development suitable for a dashboard-style application.

Backend: Python with FastAPI — chosen specifically because the system's core AI feature depends on MediaPipe and OpenCV, which are native Python libraries. Using Python end-to-end for the backend avoids running a separate service in another language just for the AI/Computer Vision component, and FastAPI provides fast, asynchronous request handling with simple integration into the same codebase as the AI logic.

2.2 Database & Servers

PostgreSQL Or MySQL will be used as the relational database for storing user accounts, fitness profiles, workout/nutrition plans, exercise performance, form-analysis results, and progress records. The frontend and backend communicate through RESTful APIs exposed by the FastAPI backend (e.g., POST /auth/login, POST /form-analysis/photo, GET /progress/{user_id}).

2.3 AI Model & Services

Pose detection: MediaPipe Pose, a pre-trained Computer Vision/deep-learning model, is used to extract 33 body landmarks from an uploaded exercise photo. A pre-trained model is used instead of training a custom model because building a labeled, exercise-form dataset large enough for reliable custom training is not feasible within the project's time and resource constraints.

Form classification: Rule-based joint-angle analysis is applied on top of the landmark output (e.g., elbow angle computed from the shoulder-elbow-wrist landmarks) to classify form as correct or incorrect against defined angle thresholds per exercise. This approach was chosen over a custom-trained ML classifier because it does not require a large labeled dataset, is deterministic and explainable, and can be reliably implemented and tested within the project timeline.

This will be served through Python FastAPI endpoints within the same backend used for the rest of the application.

3. User Classes and Characteristics

User Class	Characteristics / Permissions
End User	Registers and logs in; manages their own fitness profile; receives personalized workout/nutrition plans; uploads exercise photos for form analysis; views form feedback and score; tracks their own progress history and adaptive plan updates.
System Administrator	In this project's scope, the System Administrator role is fulfilled by the project team directly through the backend/database (not a separate admin UI). Covers managing user accounts and the exercise library/instructions; does not access other users' personal health data beyond what is required for administration.

4. Requirement Elicitation

4.1 Elicitation Techniques

We conducted a one-on-one interview with **Trainer Lion** an experienced fitness trainer at **Rayhan Fitness Mirpur** on 9 October 2026. The interview was held in person. A video of the main interview was recorded (about 1.5 minutes), and the trainer gave us further advice in an informal discussion outside the recording. We asked about many topics such as home workouts, workouts when only dumbbells are available at home, daily protein intake, and how daily calorie needs should be decided.

We also collected a real mass-gain diet plan and workout plan from a gym member who follows a mass-gain routine on that day. We reviewed it to understand what a real plan contains, such as meal structure, protein and calorie amounts, exercises, sets, reps, and weekly schedule.

4.2 Elicitation Evidence

https://drive.google.com/drive/folders/15zr5GKvVuKen2kMP2F-cfejSm-ScZ8Cj





4.3 Key Findings

Home workouts: The trainer explained that a full-body workout can be done at home. Without any equipment, suitable exercises are normal push-ups, pull-ups, and bodyweight squats. If dumbbells are available, the user can also do dumbbell curls, barbell curls, dumbbell chest press, dumbbell side raises, dumbbell squats, and dumbbell overhead triceps extension.

Protein intake: The trainer advised that a normal person needs protein roughly equal to their body weight (in grams per kg). For people who exercise, the intake should be about 1.5 to 2 times their body weight.

Calorie requirement: The trainer said that daily calorie needs depend on the person's height, weight, and age, so they cannot be guessed and must be calculated from these values.

A Muscle gain diet and workout plan Of A Client For 1 Month (beginner)

Client's muscle-gain plan: The diet plan (for a client weighing 64 kg, height about 166 cm, goal: lean muscle gain) is divided into six meals per day, with exact quantities for each meal (for example oats, eggs or whey protein, rice, chicken, fish, potato, and spinach), plus supplements such as multivitamin, fish oil, and zinc. The client's beginner workout plan had six basic exercises (push-up, assisted pull-up, dumbbell shoulder press, barbell curl, tricep pressdown, bodyweight squat), mostly 4 sets of 12 reps. After one month, the trainer updated the plan to three training days grouped by muscle area (chest/shoulders/triceps, legs, and back/biceps) with six to eight exercises per day, again at 4 sets of 12 reps.





The Updated Muscle gain workout plan Of That Client After 1 Month



Name: Sazin
Weight: 64kg
Height:166cm
(Day-1)
1. machine chest press 12/4
2.Pectoral fly 12/4
3. Deckline chest press 12/4
4.Machine shoulder press 12/4
5. D.b side raise 12/4
6. Rope press down 12/4
7. Machine dips 12/4
(Day-2)
1 Barbell squat 12/4
2. Leg press 12/4
3. V. squat 12/4
4. Seated leg curl 12/4
5 Aductor 12/4
6. standing Calf raise 12/4
(Day -3)
1.High row 12/4
2. low row 12/4
3. D.y row 12/4
4. Seated cable row 12/4
5. Face pull 12/4
6. Barbell curl 12/4
7. Dumbbel curl 12/4
8. Hammer curl 12/4

Reference: Adiyat Bin Kabir (Senior trainer Of Rayhan Fitness)

4.4 Requirement Changes After Elicitation

ID	Change	Before Elicitation	After Elicitation	Source / Reason
FR-04	Modified	The system calculates BMI, daily calorie needs, and protein target from profile information (no specific protein rule defined).	Protein target is set to about 1 g per kg of body weight for non-exercising users and 1.5 to 2 g per kg for users who exercise. Calorie needs are calculated from height, weight, and age.	Trainer interview: protein intake and calorie advice
FR-05	Modified	The system creates a workout plan from goal, fitness level, equipment, and limitations.	The plan also depends on home or gym choice. With no equipment it uses bodyweight exercises (push-up, pull-up, bodyweight squat). With dumbbells it adds dumbbell exercises, giving a full-body routine at home.	Trainer interview (home and dumbbell workouts) and the client's workout plan
FR-15	Modified	The AI updates future workout recommendations using workout history, form scores, and progress data.	The updated plan also moves the user from a beginner plan to a larger plan (more training days and exercises) after a period of training, such as one month.	Client's updated plan after 1 month

5. Functional Requirements (FR)

ID	Requirement Name	Description	User	Priority
FR-01	User Registration	The system shall allow a new user to create an account using an email and password.	End User	High
FR-02	User Login	The system shall allow registered users to log in securely.	End User	High
FR-03	Fitness Profile	The system shall allow users to enter age, gender, height, weight, fitness level, goal, workout location (home or gym),available equipment, and physical limitations.	End User	High
FR-04	BMI & Calorie Calculation	The system shall calculate the user's BMI, daily calorie needs (from height, weight, and age), and protein target (about 1 g/kg for non-exercising users, 1.5 to 2 g/kg for users who exercise).	End User	High
FR-05	Initial Workout Plan	The system shall create an initial workout plan based on the user's goal, fitness level, home or gym choice, available equipment (bodyweight only or dumbbells), and limitations.	End User	High
FR-06	Nutrition Guidance	The system shall provide calorie, protein, and basic meal guidance based on the user's goal and profile. The system shall allow users to record and review their daily food intake and nutrition information.	End User	Medium
FR-07	Exercise Library	The system shall show supported exercises with instructions and basic information.	End User	Medium
FR-08	Exercise Form Analysis	The system shall analyze an uploaded exercise photo using MediaPipe Pose and joint-angle analysis.	End User	High
FR-09	Form Score & Feedback	The system shall show a form score and explain the detected form problems to the user.	End User	High

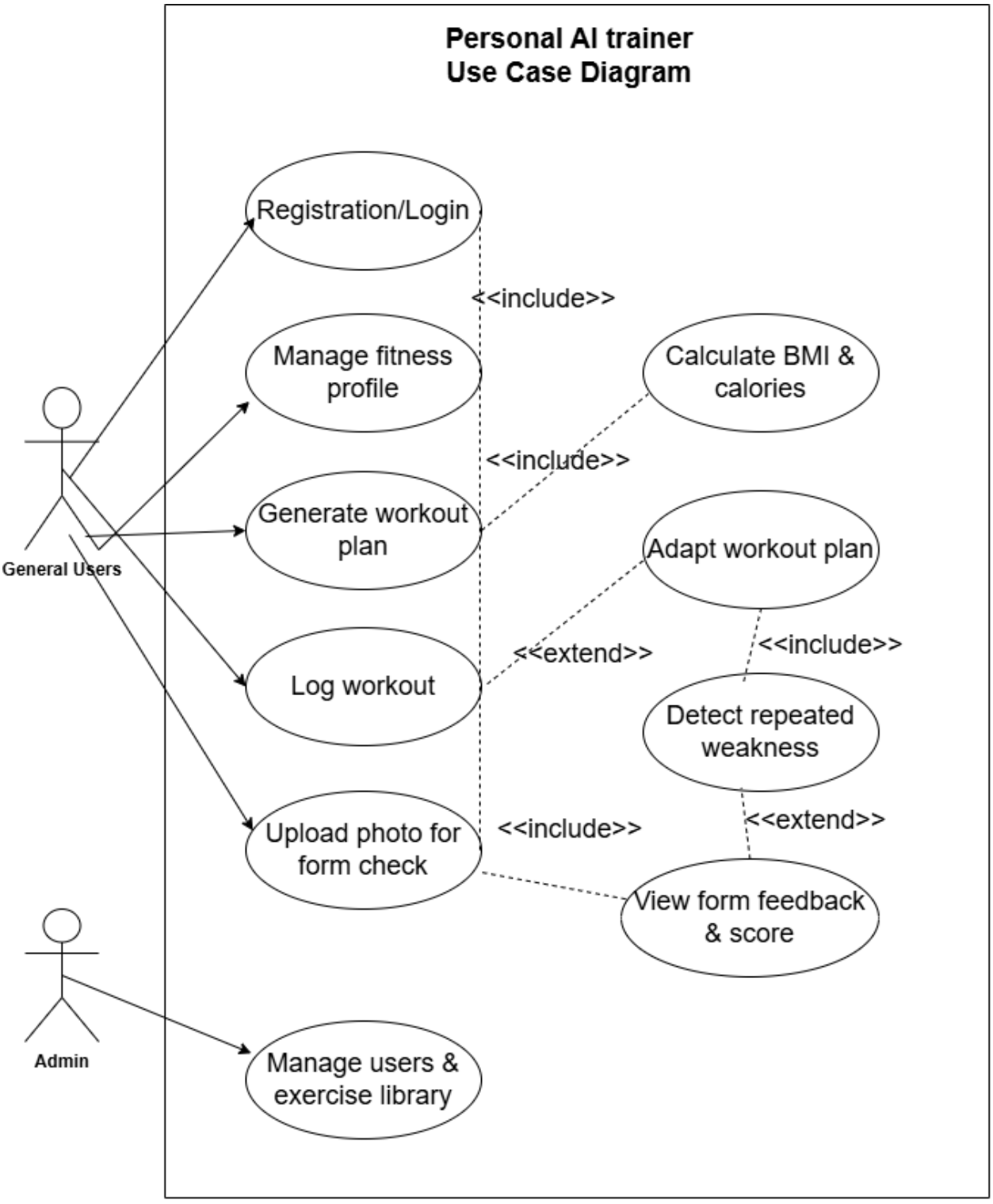
ID	Requirement Name	Description	User	Priority
FR-10	Video Form Analysis	Where possible, the system shall analyze an uploaded exercise video and check the user's form.	End User	Low
FR-11	Live Camera Analysis	Where possible, the system shall analyze exercise form using the device camera in real time.	End User	Low
FR-12	Workout Logging	The system shall allow users to record exercises, sets, reps, and weights after completing a workout.	End User	High
FR-13	Progress Tracking	The system shall show the user's weight, workout history, and form-score progress using charts.	End User	Medium
FR-14	Performance Comparison	The system shall compare the user's current performance with previous performance.	End User	Medium
FR-15	Adaptive Workout Recommendation	The AI shall use workout history, form scores, and progress data to update future workout recommendations, including moving the user from a beginner plan to a larger plan after a period of training.	End User	High
FR-16	Weakness Detection	The system shall identify repeated form problems or performance weaknesses from the user's workout data.	End User	High
sFR-17	Recommendation Explanation	The system shall explain why a specific workout or exercise recommendation was given.	End User	Medium
FR-18	User Data Management	The system shall allow users to update their profile and manage their own workout and progress data.	End User	Medium
FR-19	Exercise Library Management	The project team acting as administrator shall be able to add, update, or remove exercise information through the backend/database.	System Admin	Low

6. Non-Functional Requirements

Category	Requirement
Performance	Normal pages and requests should respond within 3 seconds under normal conditions. (photo-based form analysis may take longer due to model inference).
Security	User passwords shall be stored securely, and protected features shall require authentication
Usability	The system shall have a simple and user-friendly interface for beginners.
AI Reliability	If the system cannot detect the user's body correctly, it shall show an error message instead of giving a wrong result.
Maintainability	The frontend, backend, database, and AI components shall be organized separately so they can be maintained easily.
Privacy	Users shall only be able to access their own personal and workout data.

7. System Design Diagrams

7.1 Use Case Diagram



7.2 Use Case Description — High Priority Features

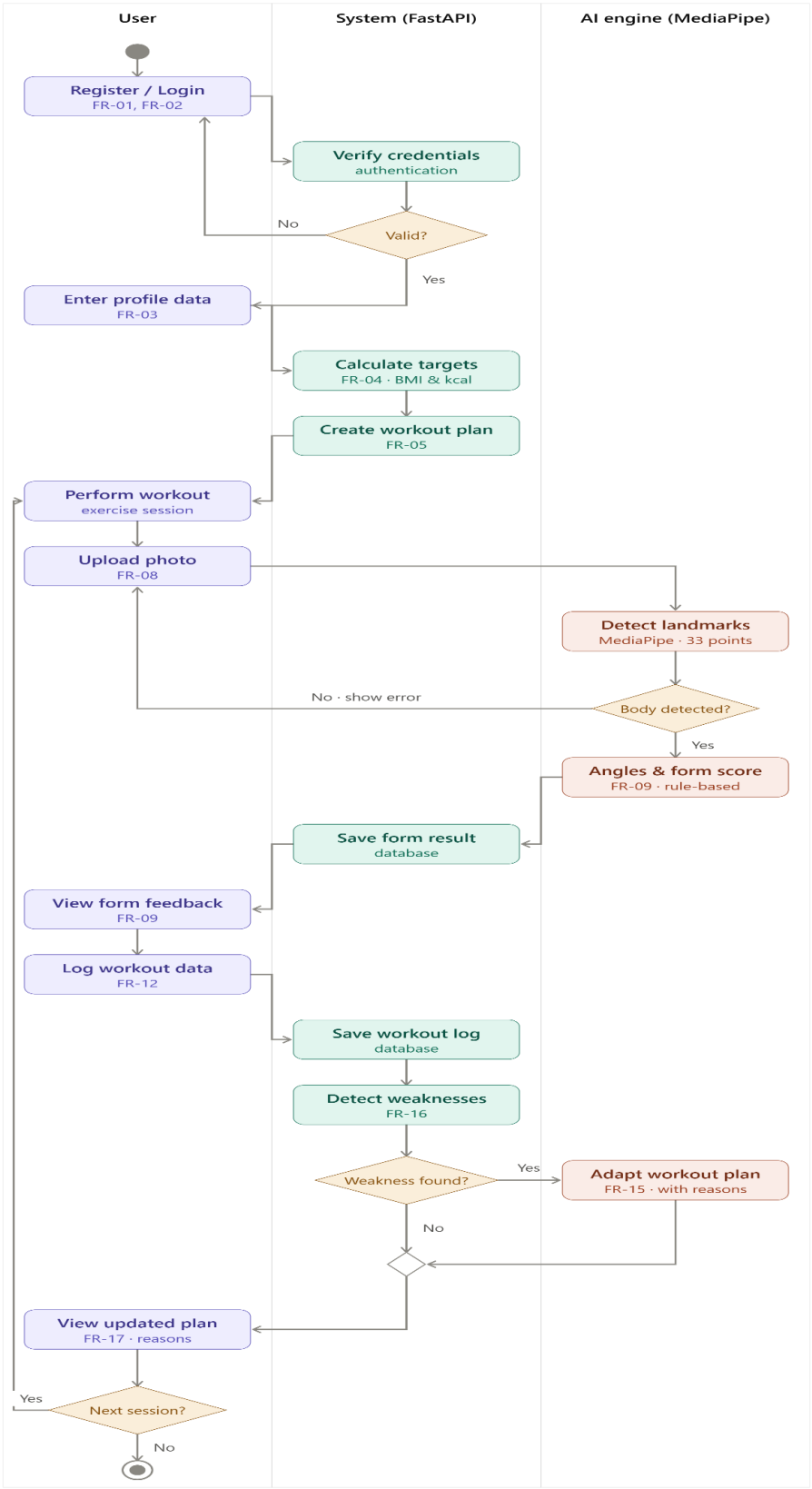
Use Case: Photo-Based Form Detection (FR-08)

Field	Description
Actor	End User
Precondition	User is logged in and has selected an exercise type.
Main Success Flow	1. User uploads an exercise photo. 2. System runs MediaPipe pose estimation on the photo. 3. System extracts body landmarks. 4. System calculates relevant joint angles. 5. System compares angles against exercise-specific thresholds. 6. System classifies the form as correct or incorrect. 7. System displays the result, score, and feedback to the user. 8. System saves the result to the database.
Alternate Flow	If landmarks cannot be detected (e.g., body not fully visible), the system shows an error and asks the user to re-upload a clearer photo.
Postcondition	A form-analysis record is saved and visible in the user's progress history.

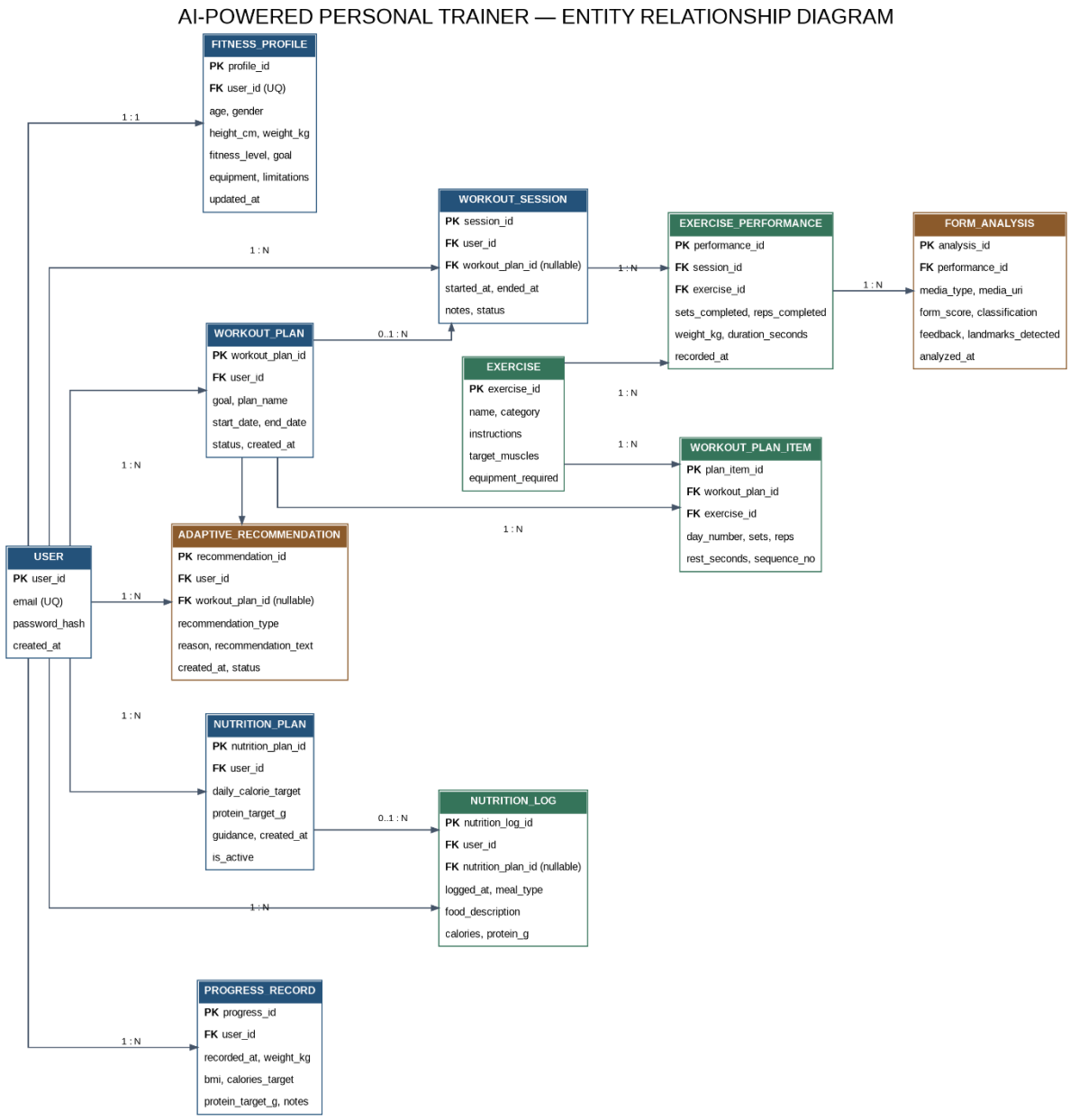
Use Case: Personalized Workout Plan Generation (FR-05)

Field	Description
Actor	End User
Precondition	User has completed their fitness profile (goal, experience, equipment, limitations).
Main Success Flow	1. User submits/updates their fitness profile. 2. System calculates BMI and calorie/protein targets. 3. System applies rule-based logic to select suitable exercises and workout frequency based on goal, level, and equipment. 4. System generates a personalized workout plan. 5. System displays the plan to the user.
Alternate Flow	If required profile fields are missing, the system prompts the user to complete them before generating a plan.
Postcondition	A workout plan is generated and stored, available for the user to follow and log sessions against.

7.3 Activity Diagram — Exercise Form Detection



7.4 Entity Relationship Diagram (ERD)



Relationship summary: USER has one FITNESS_PROFILE (1:1) and may have multiple WORKOUT_PLAN, WORKOUT_SESSION, NUTRITION_PLAN, NUTRITION_LOG, PROGRESS_RECORD, and ADAPTIVE_RECOMMENDATION records (1:N). A WORKOUT_PLAN contains multiple WORKOUT_PLAN_ITEM rows, each connecting one plan to one EXERCISE; this resolves the many-to-many relationship between plans and exercises. A WORKOUT_SESSION may reference a plan and contains multiple EXERCISE_PERFORMANCE records. Each performance record refers to one EXERCISE and may have multiple FORM_ANALYSIS records so repeated photo/video/live checks can be retained. NUTRITION_LOG belongs to one user and may optionally refer to a NUTRITION_PLAN. An ADAPTIVE_RECOMMENDATION belongs to one user and may optionally reference the workout plan it recommends. PK denotes primary key, FK denotes foreign key, and UQ denotes a unique constraint. Store media as a protected URI/path rather than embedding image/video binaries in the relational database.

7.5 AI Feature Design

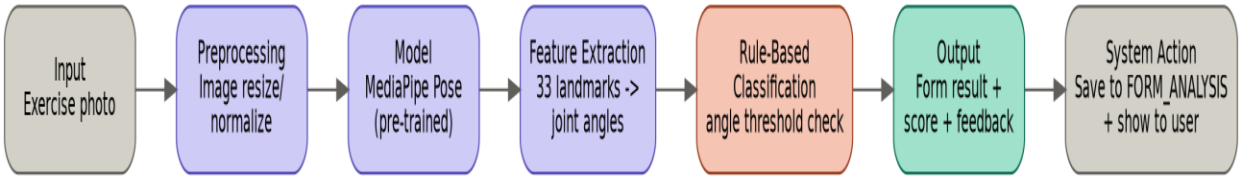
Data

Source: No custom training dataset is required for the core photo-based detection feature, since it uses the pre-trained MediaPipe Pose model for landmark detection. Joint-angle thresholds per exercise will be defined using sports-science/biomechanics references and validated against a small set of self-recorded test photos (correct and incorrect form) for each of the supported exercises (Squat, Dumbbell Curl, Push-up, Shoulder Press, Lateral Raise, Lunge, Plank).

Model

Approach: Computer Vision (MediaPipe Pose, pre-trained) for landmark extraction, combined with a rule-based classifier (joint-angle thresholds) for form classification. This is compared against the baseline alternative of training a custom ML classifier (e.g., Random Forest/SVM on landmark features), which was not chosen as the primary approach due to the lack of a large labeled dataset and limited project timeline; it remains a possible future-work extension.

Pipeline



Integration

The frontend sends the uploaded photo to a FastAPI endpoint (e.g., POST /form-analysis/photo). The backend runs the MediaPipe pipeline synchronously within the request, computes the result, stores it in the Form_Analysis table linked to the corresponding Exercise_Performance record, and returns the result to the frontend for display.

Evaluation & Fallback

The team will manually review classification results against a small set of labeled test photos per exercise and report basic accuracy (correct classifications / total test photos) before the demo. If MediaPipe fails to detect landmarks in a given photo (e.g., poor lighting or incomplete body visibility), the system will return a clear error message rather than an unreliable result, instead of a numeric fallback score.

7.5.1 Supporting AI Features

The project will implement a small set of practical AI-assisted features using the existing React/Vite frontend, FastAPI backend, and PostgreSQL database. The core computer-vision feature remains exercise-form analysis. Supporting features will use transparent rules and lightweight statistical methods so they can be implemented and tested within the project timeline.

1. Personalized Workout Generation

Input: fitness goal, experience level, available equipment, training days, and user-reported limitations. Method: a Python rule-based selector matches eligible exercises from the exercise library and assigns sets, repetitions, and a weekly schedule. Output: a stored personalized workout plan. Evaluation: test representative profile combinations and verify that equipment and limitation constraints are respected.

2. Personalized Nutrition Recommendation

Input: profile data, fitness goal, dietary preferences, and a curated nutrition database. Method: calculate estimated calorie and protein targets and recommend matching foods using transparent filtering rules. Output: daily targets and meal suggestions. Evaluation: verify calculations with test cases and ensure that missing or invalid profile values produce a clear prompt. Results are general fitness guidance, not medical advice.

3. Workout Weakness Detection

Input: workout sessions, exercise performance, and saved form-analysis results. Method: analyze repeated low form scores, recurring technique flags, missed sessions, or declining recorded performance using configurable thresholds. Output: a ranked list of observed training weaknesses with the records that support each finding. Evaluation: use seeded test histories to check whether known patterns are detected.

4. Adaptive Workout Recommendation

Input: recent workout history, completed sets/repetitions/weights, form scores, and user-reported feedback. Method: conservative rules decide whether to maintain, slightly increase, or reduce a planned exercise workload; poor form or reported pain prevents an automatic increase. Output: a proposed plan adjustment with a short explanation. The user remains in control of accepting changes.

5. Progress Trend Prediction

Input: dated historical measurements and workout performance. Method: use a lightweight scikit-learn Linear Regression model only when sufficient valid records exist; otherwise show historical trends without a prediction. Output: an estimated short-term trend and chart, clearly labelled as an estimate rather than a guarantee. Evaluation: compare predictions with held-out historical records and report an appropriate error metric.

6. Explainable Form Feedback

Input: detected body landmarks, calculated joint angles, exercise-specific thresholds, and classification results. Method: map each triggered rule to a reviewed feedback template. Output: an explanation of the detected issue and a technique cue linked to the analysis result. If body landmarks are unreliable, the system must request another image instead of inventing feedback.

7.5.2 Integration and Data Flow

The React frontend sends authenticated requests to FastAPI. The backend validates inputs, retrieves the user's authorized records from PostgreSQL, executes the applicable vision pipeline, rule engine, or statistical model, and stores generated plans, analysis results, and recommendations. Each saved result should include its creation time and relevant source records so the output can be explained and audited. Uploaded photos and videos should be stored in protected file/object storage; the database should retain only a protected media URI and metadata.

7.5.3 Evaluation, Reliability, and Scope

The team will test each feature using documented normal, boundary, and invalid-input cases. Form analysis will be evaluated against a small manually labelled test set, reporting correct classifications divided by total evaluated cases. Recommendation features will be tested with predefined profiles and workout histories. Progress prediction will be enabled only when enough historical data is available. The system will not diagnose medical conditions, prescribe treatment, or claim that heuristic rules are a trained machine-learning model. All recommendations are advisory, and uncertain or insufficient data must result in a clear limitation message.

8. Team Member Responsibility Matrix

Member Name	ID	Primary Responsibilities
Asif Karim Alif	241-35-003	AI/ML: pose estimation integration, joint-angle rule logic, form-analysis pipeline, and evaluation.
MD. Mojammel Nahid	241-35-206	Frontend UI/UX, dashboard, and state management (React/Vite).
Md Arik Abrar	241-35-163	Backend API development, database schema, and authentication (FastAPI, PostgreSQL).

Note: While each member holds primary ownership of one area above for individual evaluation purposes, the team follows a collaborative workflow where all members participate in requirement analysis, integration, testing, and documentation across the full system, as outlined in the project proposal.